MATH 189 - Data Analysis

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Contents

1 Statistical Learning

1.1 Terminology

"Supervised Learning" means we have both predictors and responses.
"Predictors/input variables/features" are arranged in a vector.

$$X = (X_1, X_2, ..., X_n)$$

"Output variables/responses" is a scalar where $y \in \mathbb{R}$
We think of data as coming from a model.

$$y = f(x) + \epsilon$$

$$\begin{cases} f(x) \text{ is fixed but unknown function} \\ \epsilon \text{ is the random error term with mean zero, independent of X} \end{cases}$$

Question: Why estimate f?
Prediction: Given X, we want to predict y with $\hat{y} = \hat{f}(x)$

Mean square error is

$$\begin{aligned} \mathbb{E}(y - \hat{y})^2 &= \mathbb{E}(f(x) + \epsilon - \hat{f}(x)) \\ &= \mathbb{E}(f(x) - \hat{f}(x))^2 + 2\mathbb{E}((f(x) - \hat{f}(x))\epsilon) + \mathbb{E}(\epsilon^2) \end{aligned}$$

Examine the middle term in detail

$$2\mathbb{E}((f(x) - \hat{f}(x))\epsilon) = \mathbb{E}(f(x) - \hat{f}(x)) \cdot \mathbb{E}(\epsilon) = 0$$

We know that

$$\mathbb{E}(y - \hat{y})^2 = (f(x) - \hat{f}(x))^2 + \mathbb{E}(\epsilon^2)$$

Where the left term is reducible error and the right term is irreducible error.

"Inference" - Sometimes we want to understand the association between y and $X_1, ..., X_p$

1. Which predictors are associated with response?

2. Is the association positive or negative?

3. Is the relationship approximately linear?

1.2 Accuracy

Training data

$$(x_1, y_1), (x_2, y_2), ...(x_n, y_n)$$
$$\text{where } X_i = (X_{i1}, X_{i2}, ..., X_{ip})$$

n = # data points (e.g. $n = 30$)
p = # predictors (e.g. $p = 3$)

Mean Square Error (MSE)

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{f}(x_i))^2$$
$$\text{Defined on training data}$$
$$\text{Test MSE} = \text{ Ave } (y_0 - \hat{f}(x_0))^2$$
$$\text{Defined on previously unseen data point } (x_0, y_0)$$

Linear / affine model

$$f(x) = \beta_0 + \beta_1 x_1 + ... + \beta_p x_p$$

"Fit/train the model" usually means run an optimization on your computer that uses training data and outputs the parameters $\beta_0, \beta_1, ..., \beta_p$ that make the MSE and hopefully the test MSE small.

Classification
Data set $(x_1, y_1), (x_2, y_2,), ..., (x_n, y_n)$ where $X_i = (x_{i1}, ..., x_{ip})$ and y_i is qualitative.

Error rate is the fraction of incorrect classifications.

$$\frac{1}{n}\sum_{i=1}^{n}\mathbb{I}\{y_i \neq \hat{y}_i\}$$

$$\mathbb{I}\{y_i \neq \hat{y}_i\} = \begin{cases} 1, & y_i \neq \hat{y}_i \\ 0, & y_i = \hat{y}_i \end{cases}$$

Test error rate

$$\text{Ave}(\mathbb{I}\{y_0 \neq \hat{y}_0\})$$

Where (x_0, y_0) is a previously unseen test data point.

K-nearest neighbors (KNN)
The KNN classifier is defined by letting η_0 = set of indices $i \in \{1, ..., n\}$ for K nearest neighbors of X_0 among the input data.

$\hat{y}_0 = j$ where j maximizes

$$\sum_{i \in \eta_0} \mathbb{I}\{y_i = j\}$$

which is the number of nearest neighbors (out of K) assigned label j.

2 Linear Regression

2.1 Simple Linear Regression

Underlying statistical model: $y = \beta_0 + \beta_1 X + \epsilon$ where y is the quantitative response and a single predictor X which is why it is called simple.

Prediction model: $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 X$
$\hat{\beta}_0, \hat{\beta}_1$ are estimates based on training data, where hat means estimated quantity.

Estimating coefficients
The residual sum of squares (RSS)

$$\begin{aligned} \text{RSS} &= \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \\ &= \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 \end{aligned}$$

Choose $\hat{\beta}_0, \hat{\beta}_1$ to minimize RSS.

$$\begin{aligned}
\frac{\partial \text{RSS}}{\partial \hat{\beta}_0} &= \frac{\partial}{\partial \hat{\beta}_0} \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 X)^2 \\
&= 2 \sum_{i=1}^{n} (\hat{\beta}_0 + \hat{\beta}_1 X_i - Y_i) \\
&\implies \text{RSS is } \begin{cases} \text{decreasing for really small } \hat{\beta}_0 \\ \text{increasing for really big } \hat{\beta}_0 \end{cases}
\end{aligned}$$

RSS is optimal (in $\hat{\beta}_0$) when

$$\begin{aligned}
0 &= \frac{1}{n} \sum_{i=1}^{n} (\hat{\beta}_0 + \hat{\beta}_1 X_i - Y_i) \\
&= \hat{\beta}_0 + \hat{\beta}_1 \Big(\frac{1}{n} \sum_{i=1}^{n} X_i \Big) - \Big(\frac{1}{n} \sum_{i=1}^{n} Y_i \Big) \\
&= \hat{\beta}_0 + \hat{\beta} \bar{x} - \bar{y} \\
\hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x}
\end{aligned}$$

Where $\begin{cases} \bar{x} = \frac{1}{n} \sum_{i=1}^{n} X_i \text{ is the mean predictor} \\ \bar{y} = \frac{1}{n} \sum_{i=1}^{n} Y_i \text{ is the mean response} \end{cases}$

<u>Covariance</u>

$$\begin{aligned}
\text{Cov}(X_i, Y_i)_{i=1}^{n} &= \frac{1}{n} \sum_{i=1}^{n} X_i Y_i - \bar{X}\bar{Y} \\
&= \frac{1}{n} \sum_{i=1}^{n} (X_i - \bar{x})(Y_i - \bar{y})
\end{aligned}$$

<u>Variance</u>

$$\begin{aligned}
\text{Var}(X_i)_{i=1}^{n} &= \text{Cov}(X_i, X_i)_{i=1}^{n} \\
&= \frac{1}{n} \sum_{i=1}^{n} (X_i - \bar{x})^2
\end{aligned}$$

From this we can get our optimal $\hat{\beta}_1$

$$
\begin{aligned}
\hat{\beta}_1 &= \frac{\mathrm{Cov}(X_i, Y_i)_{i=1}^n}{\mathrm{Var}(X_i)_{i=1}^n} \\
&= \frac{\frac{1}{n}\sum_{i=1}^n (X_i - \bar{x})(Y_i - \bar{y})}{\frac{1}{n}\sum_{i=1}^n (X_i - \bar{x})^2}
\end{aligned}
$$

2.2 Goodness of Fit & Correlation

Assume an underlying statistical model

$$y = \beta_0 + \beta_1 X + \epsilon$$

We use training data $(X_i, Y_i)_{i=1}^n$ to generate prediction model $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 X$
The formulas for $\hat{\beta}_0$ and $\hat{\beta}_1$ to minimize RSS:

$$\mathrm{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \begin{cases} \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{X} \\ \hat{\beta}_1 = \frac{\mathrm{Cov}(X_i, Y_i)_{i=1}^n}{\mathrm{Var}(X_i)_{i=1}^n} \end{cases}$$

Goodness of fit

$$
\begin{aligned}
\mathrm{RSS} &= \sum_{i=1}^n (y_i - \hat{y}_i)^2 \\
&= \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 \\
&= \sum_{i=1}^n \Big((y_i - \bar{y}) - \hat{\beta}_1 (x_i - \bar{x}) \Big)^2 \\
&= \sum_{i=1}^n (y_i - \bar{y})^2 - 2\hat{\beta}_1 \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) + \hat{\beta}_1^2 \sum_{i=1}^n (x_i - \bar{x})^2 \\
&= \sum_{i=1}^n (y_i - \bar{y})^2 - \frac{\Big[\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})\Big]^2}{\sum_{i=1}^n (x_i - \bar{x})^2}
\end{aligned}
$$

RSS scales with the number of data points n and also scales with y^2 units.
$\rightarrow$ We can fix the scaling and get out a goodness-of-fit score (R^2) between 0 and 1.

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}}$$

$$\text{RSS} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \qquad \text{TSS} = \sum_{i=1}^{n} (y_i - \bar{y})^2$$

So for simple linear regression,

$$\begin{aligned} R^2 &= 1 - \frac{\text{RSS}}{\text{TSS}} \\ &= \frac{\text{TSS} - \text{RSS}}{\text{TSS}} \\ &= \frac{\left[\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})\right]^2}{\sum_{i=1}^{n} (x_i - \bar{x})^2 \sum_{i=1}^{n} (y_i - \bar{y})^2} \\ R &= \frac{\text{Cov}((x_i), (y_i))}{\text{Var}(x_i)^{1/2} \text{Var}(y_i)^{1/2}} \end{aligned}$$

Long story short R^2 is the fraction of variance explained by the model.

$\boxed{ex}$ $\widehat{\text{sales}} = 7.0 + 0.048 \cdot \text{TV}$
$R^2 = 0.61 = 61\%$ $r \approx 0.78$ pretty strong positive correlation

2.3 Regression Table

$\boxed{ex}$

	Coefficient	Std. error	t-statistic	p-value
Intercept (β_0)	7.0	0.46	15	<0.0001
TV (β_1)	0.048	0.0027	18	<0.0001

Std error is the error bar on linear coefficients.

$$\text{We know that} \begin{cases} \beta_0 \approx \hat{\beta}_0 \pm \text{SE}(\hat{\beta}_0) \\ \beta_1 \approx \hat{\beta}_1 \pm \text{SE}(\hat{\beta}_1) \end{cases}$$

Assume the underlying statistical model is true!

$$y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

where ϵ_i are independent $\eta(0, \sigma^2)$ random variables

If we generate N data points (x_i, y_i) from the model, then

$$\begin{cases} \text{Var}(\hat{\beta}_0)^{1/2} \approx \text{Std. }\hat{\ }\text{error}(\hat{\beta}_0) \\ \text{Var}(\hat{\beta}_1)^{1/2} \approx \text{Std. }\hat{\ }\text{error}(\hat{\beta}_1) \\ \mathbb{E}(\hat{\beta}_0) = \beta_0 \\ \mathbb{E}(\hat{\beta}_1) = \beta_1 \end{cases}$$

2.4 Multiple Linear Regression

The underlying statistical model that we assume is true.

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + ... + \beta_p x_p$$

where y is response, β_0 is the intercept, $\beta_1 x_1, ..., \beta_p x_p$ are predictors, and ϵ is the mean zero noise.

$\boxed{ex}$ $\quad \text{sales} = \beta_0 + \beta_1 \text{TV} + \beta_2 \text{Radio} + \beta_3 \text{Newspaper} + \epsilon$

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + ... + \hat{\beta}_p x_p$$

We want to find the best $\hat{\beta}$ to minimize RSS, where $\hat{\beta}$ are estimated quantities.

$$\begin{aligned} RSS &= \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \\ &= \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - ... - \hat{\beta} x_{ip})^2 \\ &= \left|\left| \begin{bmatrix} y_1 - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - ... - \hat{\beta}_p x_{ip} \\ \vdots \\ y_n - \hat{\beta}_0 - \hat{\beta}_1 x_{n1} - ... - \hat{\beta}_p x_{ip} \end{bmatrix} \right|\right|^2 \\ &= ||Y - X\hat{\beta}||^2 \end{aligned}$$

where $y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \qquad \hat{\beta} = \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \vdots \\ \hat{\beta}_p \end{bmatrix} \qquad X = \begin{bmatrix} 1 & x_{11} & \dots & x_{1p} \\ 1 & x_{21} & \dots & x_{1p} \\ & & \ddots & \\ 1 & x_{n1} & \dots & x_{np} \end{bmatrix}$

The best $\hat{\beta}$ to minimize Euclidean norm^2 for $||Y - X\hat{\beta}||^2$ is $\hat{\beta} = X^+Y = (X^TX)^{-1}X^TY$

	Coefficient	Std. error	t-statistic	p-value
Intercept	$\hat{\beta}_0$ 2.939	0.3119	9.42	<0.0001
TV	$\hat{\beta}_1$ 0.046	0.0014	32.81	<0.0001
Radio	$\hat{\beta}_2$ 0.189	0.0086	32.89	<0.0001
Newspaper	$\hat{\beta}_3$ -0.001	0.0059	-0.18	0.8599

Note: the standard error is NOT $Var(\hat{\beta}_i)^2$ but rather $\hat{Var}(\hat{\beta}_i)^2$

<u>Hypothesis testing</u>

State a hypothesis, test it with data (testing the likelihood such data could possibly be generated if the hypothesis were true).

H_0 - the null hypothesis: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \epsilon$ for some coefficients B_0, β_2, β_3 with $\beta_1 = 0$ with $\epsilon \sim N(0, \sigma^2)$ for some variance $\sigma^2 \geq 0$. Also, observations are independent.

H_1 - the alternative hypothesis: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \epsilon$ where everything above holds except $\beta_1 \neq 0$

Assume H_0. Then

$$\frac{\hat{\beta}_1}{\hat{SE}(\hat{\beta}_1)} = t \sim T_{n-p-1}$$

3 Logistic Regression & Generative Models

3.1 Logistic Regression

We want to predict whether an event happens ($y = 1$) or not ($y = 0$), we really want to model the probability $p \in (0, 1)$ of occurrence $\implies$ NOT LINEAR REGRESSION.

<u>Before</u>

$$y = \beta_0 + \beta_1 X_\epsilon$$

Now

$$p = \frac{\exp(\beta_0 + \beta_1 X)}{1 + \exp(\beta_0 + \beta_1 X)}$$
$$1 - p = \frac{1}{1 + \exp(\beta_0 + \beta_1 X)}$$
$$\frac{p}{1-p} = \exp(\beta_0 + \beta_1 X)$$

Where the term $\frac{p}{1-p}$ is called "odds".
Note: $\log(\text{odds}) = \beta_0 + \beta_1 X$

$\boxed{ex}$ Defaulting on bank loans

$$p = \text{probability of defaulting}$$
$$X = \text{balance}$$
$$y_i = \begin{cases} 0 & \text{default doesn't happen} \\ 1 & \text{default does happen} \end{cases}$$
$$\log(\frac{p}{1-p}) = \beta_0 + \beta_1 \cdot \mathrm{X}$$

How do we use data/observations $(x_i, y_i)_{1 \le i \le n}$ to bestimate β_0, β_1.

$$\begin{cases} \text{Linear regression} \implies \text{minimize RSS} \\ \text{Logistic regression} \implies \text{use max likelihood, do it on a computer} \\ \text{Generative models} \implies \text{simple formulas for } \hat{\Pi}_k, \hat{\mu}_k, \hat{\Sigma}_k \end{cases}$$

We choose estimates $\hat{\beta}_0, \hat{\beta}_1$ to maximize likelihood of observations $(x_i, y_i)_{1 \le i \le n}$ if (a) the observations were independent and (b) they were generated exactly from the statistical model.

$$\begin{aligned} \mathbb{P}\left((x_i, y_i)_{1 \le i \le n}\right) &= \prod_{i=1}^{n} \mathbb{P}(x_i, y_i) \text{ where } y_i \in \{0, 1\} \\ &= \prod_{y_i=1} \hat{p}(x_i) \prod_{y_i=0} (1 - \hat{p}(x_i)) \\ &= \prod_{i:y_i=1} \frac{\exp(\beta_0 + \beta_1 X_i)}{1 + \exp(\beta_0 + \beta_1 X_i)} \prod_{i:y_i=0} \frac{1}{1 + \exp(\beta_0 + \beta_1 X_i)} \end{aligned}$$

Know that $\hat{\beta}_0, \hat{\beta}_1$ are chosen to maximize this probability/likelihood above.

$\boxed{ex}$ $P_{\text{defaulting}}$

	Coefficient	Std. error	z-statistic	p-value
Intercept	-10.6513	0.3612	-29.5	<0.0001
Balance	.0055	0.0002	24.9	<0.0001

We want $\hat{\beta}_0 + \hat{\beta}_1 X = 0$

$$\begin{aligned} -10.6513 + 0.0055X &= 0 \\ X &= \frac{10.6513}{0.0055} \\ &= 1936.6 \end{aligned}$$

Assume

a. The model is true

b. Observations $(x_i, y_i)_{1 \le i \le n}$ are independent draws from the model

c. Model is true with $\beta_0 = 0$ (null hypothesis)

3.2 Generative Models for Classification

Predictors
X or $X_1, X_2, ..., X_n$
Response
$y \in \{0, 1\}$ or $y \in \{1, 2, ..., k\}$

Review
Simple logistic regression

$$P = \text{prob}\{y = 1\} = \frac{\exp(\beta_0 + \beta_1 X)}{1 + \exp(\beta_0 + \beta_1 X)}$$

Note: This doesn't make any assumptions about how predictors are distributed.

For each $k = 1, 2, ..., K$:
Our statistical model ('mixture of Gaussians') says $y = k$ with probability π_k ($\sum_{k=1}^{K} \pi_k = 1$)
Given $y = k$, $X \sim N(\mu_k, \Sigma_k)$ where Σ is the covariance matrix symbol. Here is the general vector-valued format:

$$X = (X_1, ..., X - p) \qquad \Sigma_k = \begin{bmatrix} & \end{bmatrix} \text{pxp matrix}$$

Where μ_k has Kp parameters and Σ_k has $K\frac{p(p+1)}{2}$ parameters
When $p = 1$, we can write

$$X \sim N(\mu_k, \sigma_k^2)$$

In approaches like Linear Discriminant Analysis (LDA), Quadratic Discriminant Analysis (QDA), and Naive Bayes (NB) use simple estimators for Π_k, μ_k, Σ_k that we can calculate fast.

$\boxed{ex}$ Linear discriminant analysis (LDA) make assumptions of common covariance structure $\Sigma = \Sigma_1 = \Sigma_2 = ... = \Sigma_k$ which reduces a lot of complexity. Given observations $(X_i, y_i)_{1 \leq i \leq n}$, we can estimate for $k = 1, ..., K$

$$\begin{aligned}
\hat{\Sigma} &= \frac{n_k}{n}, \text{ where} \\
n &= \#\{i : y_i = k\} \\
&= \text{number of observations in class } k \\
\hat{\mu}_k &= \frac{1}{n_k} \sum_{i:y_i=k} X_i
\end{aligned}$$

Last estimate common covariance by

$$\hat{\Sigma} = \frac{1}{n-k} \sum_{k=1}^{K} \sum_{i:y_i=k} (X_i - \mu_k)(X_i - \mu_k)^T$$
$$\mathbb{E}\hat{\Sigma} = \Sigma \text{ if the model is true}$$

$\boxed{ex}$ Quadratic discriminant analysis is pretty much the same except DIFFERENT covariances $\Sigma_1, ..., \Sigma_k$ for each class.
$\implies \hat{\Pi}_k = \frac{n_k}{n}$, $\hat{\mu}_k = \frac{1}{n_k} \sum_{i:y_i=k} X_i$

$\implies \hat{\Sigma}_k = \frac{1}{n_k - 1} \sum_{i:y_i=k} (X_i - \mu_k)(X_i - \mu_k)^T$
$\implies K\frac{p(p+1)}{2}$ covariance free parameters.
$\boxed{ex}$ Naive Bayes is pretty much the same except DIFFERENT covariances $\Sigma_1, ..., \Sigma_k$ that are also diagonal matrices.
$\implies Kp$ covariance free parameters with the same estimators as QDA.

$$\text{diag}\left(\sum_k^n\right) = \frac{1}{n_k - 1} \sum_{i:y_i=k} (X_i - \mu_k) \odot (X_i - \mu_k)$$

Suppose you have the statistical model with probability Π_k, we set $y = k$ and draw $X \sim N(\mu_k, \Sigma_k)$. Suppose we have estimated the parameters well - see a new predictor vector $X = (X_1, ..., X_p)$ - How do you determine class probabilities $P\{y = k|X\}$ and make predictions?

Prediction part is easy - just predict based on the highest probability class: $\hat{y} = k$ for $k = \text{argmax}_{1 \leq j \leq k}\ P\{y = j|X\}$

So what is $P\{y = k|X\}$? Bayes' rule tells us

$$\begin{aligned} P\{y = k|X\} &= \frac{P\{x|y = k\}\ P\{y = k\}}{\sum_{j=1}^k P\{X|y = j\}P\{y = j\}} \\ &= P(X) \\ &= \frac{f_k(X)\Pi_k}{\sum_{j=1}^K f_j(X)\Pi_j} \end{aligned}$$

where Gaussian density $\Pi_j(X) = \frac{1}{(2\pi)^{p/2}(\det \Sigma)^{1/2}} \exp(\frac{-1}{2}(X - \mu_j)\Sigma_j(X - \mu_j))$

3.3 Discriminant Analysis In Detail

Let's say we have a generative statistical model with probability $\frac{1}{2}, y = 1$. Otherwise, $y = 2$.
Given $y = 1$, $X \sim N(\mu_1, \sigma^2)$
Given $y = 2$, $X \sim N(\mu_2, \sigma^2)$
We think about probability of class membership using Bayes' rule.

$$P\{Y = k|X\}$$

Where Y is response, k is class, either 1 or 2, and X is the predictor.

$$\begin{aligned}
&\frac{P\{X, Y=k\}}{P\{X\}} \\
&= \frac{P\{X, y=k\}}{\sum_{j=1}^{2} P\{X, y=j\}} \\
&= \frac{P\{y=k\}P\{X|y=k\}}{\sum_{j=1}^{2} P\{y=j\}P\{X|y=j\}}
\end{aligned}$$

Plugging in the specific model for class 1,

$$\begin{aligned}
P\{y=1|X\} &= \frac{\frac{1}{2}f_1(X)}{\frac{1}{2}f_1(X)+\frac{1}{2}f_2(X)} \\
\text{where } f_1(x) &= P\{X|y=1\} \\
&= \frac{1}{\sqrt{2\pi\sigma^2}}\exp(-\frac{1}{2\sigma^2}(X-\mu_1)^2) \\
f_2(x) &= P\{X|y=2\} \\
&= \frac{1}{\sqrt{2\pi\sigma^2}}\exp(-\frac{1}{2\sigma^2}(X-\mu_2)^2)
\end{aligned}$$

Usually our task in this class is to predict y as well as possible $\hat{y}=1$ or $\hat{y}=2$. The best we can do, the only normal strategy, is to predict the higher probability count.

$$\hat{y}=1 \text{ if } P\{y=1|X\} > P\{y=2|X\}$$

Which happens if and only if

$$\begin{aligned}
\frac{1}{\sqrt{2\pi\sigma^2}}\exp(-\frac{1}{2\sigma^2}(X-\mu_1)^2) &> \frac{1}{\sqrt{2\pi\sigma^2}}\exp(-\frac{1}{2\sigma^2}(X-\mu_2)^2) \\
\exp(-\frac{1}{2\sigma^2}(X-\mu_1)^2) &> \exp(-\frac{1}{2\sigma^2}(X-\mu_2)^2) \\
-\frac{1}{2\sigma^2}(X-\mu_1)^2 &> -\frac{1}{2\sigma^2}(X-\mu_2)^2 \\
(X-\mu_2)^2 &> (X-\mu_1)^2
\end{aligned}$$

So we make the prediction, $\begin{cases} \hat{y}=1 & \text{if } |X-\mu_1| < |X-\mu_2| \\ \hat{y}=2 & \text{if } |X-\mu_2| < |X-\mu_1| \end{cases}$

We want to move from 1 predictor X to two predictors X_1 and X_2

Before
$P\{y=1\} = P\{y=2\} = \frac{1}{2}$
Given $y = 1$, $X \sim N(\mu_1, \sigma^2)$
Given $y = 2$, $X \sim N(\mu_2, \sigma^2)$

After
$P\{y=1\} = P\{y=2\} = \frac{1}{2}$
Given $y = 1$, $X \sim N(\mu_1, \sigma^2)$
Given $y = 2$, $X \sim N(\mu_2, \sigma^2)$
Where $\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$ and $\vec{\mu}_1 = \begin{bmatrix} \mu_{11} \\ \mu_{12} \end{bmatrix}$ and $\vec{\mu}_2 = \begin{bmatrix} \mu_{21} \\ \mu_{22} \end{bmatrix}$

Making it bold to turn scalars into vectors works except with variance.

$$\Sigma = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}$$

4 Model Selection

4.1 Bootstrap

Quantitative response - "train" different multiple linear regression models with different predictors.
Qualitative response - LDA / QDA / NB multiple logistic regression models with different predictors.

To get the best model, consider:

1. Set aside validation data set

$$\begin{array}{ccc} & (X_i, y_i)_{1 \le i \le n} & \\ \swarrow & & \searrow \\ (X_i, y_i)_{i \in T} & & (X_i, y_i)_{i \in T^C} \end{array}$$

   Where T is a set of training data point indices, e.g. $T = \{1, ..., 100\}$ and T^C is the "test/validation" data points, e.g. $T^C = \{101, ..., 110\}$. $\hat{y}_i$ is the prediction for input X_i often using training data $(X_i, y_i)_{i \in T}$ to

train the model. We can measure $\text{MSE}_{\text{test}} =$
$\Rightarrow$ Choose the model with the best (lowest) MSE_{test}

2. Leave-one-out cross-validation (LOOCV)
   Idea: use all possible train-test splits.

$$(X_j, y_j)_{j \neq i}$$

3. K-fold cross validation
   Split your data according to k index sets $\{1, ..., n\} = S_1 \cup S_2 \cup ... \cup S_k$

4. Bootstrap
   $\boxed{ex}$ you invest $\begin{cases} \alpha & \text{fraction of your money in asset with return value } X \\ 1-\alpha & \text{fraction in asset with return value } y \end{cases}$
   You want to minimize variance of your return.
   Choose α to minimize

$$\text{Var}\left[\alpha X + (1-\alpha)y\right] = \text{Cov}[\alpha X + (1-\alpha)y, \alpha X + (1-\alpha)y]$$

   To optimize, set α derivative equal to zero.

$$\begin{aligned} \alpha &= \frac{\text{Cov[y-x, y]}}{\text{Var[y-x]}} \\ &= \frac{\sigma_y^2 - \sigma_{xy}}{\sigma_x^2 + \sigma_y^2 - 2\sigma_{xy}} \end{aligned}$$

   In practice, we use data $(x_i, y_i)_{1 \leq i \leq n}$ to estimate

$$\hat{\alpha} = \frac{\hat{\sigma}_y^2 - \hat{\sigma}_{xy}}{\hat{\sigma}_x^2 + \hat{\sigma}_y^2 - 2\hat{\sigma}_{xy}}$$

$$\text{where} \begin{cases} \hat{\sigma}_x^2 = \frac{1}{n-1}\sum_{i=1}^n (x_i - \bar{x})^2 \\ \hat{\sigma}_y^2 = \frac{1}{n-1}\sum_{i=1}^n (y_i - \bar{y})^2 \\ \hat{\sigma}_{xy} = \frac{1}{n-1}\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) \end{cases}$$

   Q: What is a good error bar for $\hat{\alpha}$?
   Q: what is $\text{SE}[\hat{\alpha}] = \text{Var}[\hat{\alpha}]^{\frac{1}{2}}$?
   Bootstrap answers this.

   <u>Bootstrap with $B(\approx 10^2 - 10^9)$ samples</u>
   For $j = 1, ..., B$:
   Make fake data set $(\tilde{x}_i, \tilde{y}_i)_{1 \leq i \leq n}$ that comes from real observations $(x_i, y_i)_{1 \leq i \leq n}$ by uniformly sub-sampling <u>with replacement</u>.

4.2 Linear Model Selection

Goal: Choose best linear regression model based on predictor selection and modification of predictors.

Question: Given a data set with p predictors and a quantitative response variable, which subset of predictors leads to the best multiple linear regression model?

1. Best subset selection
   - Try all 2^p models and evaluate the MSE on a validation data set.
   - Choose the best one.
2. Forward selection
   - Start with an empty set of variables.
   - Try adding each of the remaining variables and calculate the MSE on a validation set.
   - Choose the best variable and add it to the model if it improves the MSE.
   - Repeat until convergence.
3. Backward selection
   - Start with all p variables.
   - Try kicking out each of the variables and calculate MSE on a validation set.
   - If it improves MSE, make the best alteration to the model.
   - Repeat until convergence.

Computational cost. The main advantage of forward and backward selection is computational cost. They require at most $p+(p-1)+...+1 = p(p+1)/2$ models, and

$$p(p+1)/2 \ll 2^p$$

Accuracy. There are no guarantees that forward or backward selection find the best model, but sometimes we get lucky. The models discovered by different methods can be slightly different.

Ridge regression. One way to improve the coefficients $\hat{\beta}_0, ..., \hat{\beta}_p$ that arise in a linear model is by changing the fitting procedure. Instead of minimizing the residual sum of squares

$$\text{RSS} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

over the data set $(x_i, y_i)_{1 \leq i \leq n}$, we choose $\hat{\beta}_0, ..., \hat{\beta}_p$ to minimize

$$\text{RSS} + \underbrace{\lambda \sum_{j=1}^{p} \hat{\beta}_j^2}_{\text{shrinkage penalty}}$$

The shrinkage penalty keeps the coefficients $\hat{\beta}_1, ..., \hat{\beta}_p$ small in magnitude, note that we don't penalize the intercept $\hat{\beta}_0$. A couple limits are worth considering.

- When $\lambda = 0$, we get multiple linear regression.
- When $\lambda \to$, all coefficients except the intercept approach zero.

It is best to apply ridge regression (or lasso) after standardizing each predictor by subtracting the mean and dividing by the standard deviation.

$$x_{ij} \to \tilde{x}_{ij} = \frac{x_{ij} - \bar{x}_j}{\sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_{ij} - \bar{x}_j)^2}} \text{ where } \bar{x}_j = \frac{1}{n}\sum_{i=1}^{n} x_{ij} \text{ is mean of predictor } j.$$

Lasso. Instead of minimizing $\text{RSS} = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$, the lasso chooses $\hat{\beta}_1, ..., \hat{\beta}_p$ to minimize

$$\text{RSS} + \underbrace{\lambda \sum_{j=1}^{p} |\hat{\beta}_j|}_{\text{lasso penalty}}$$

Note that the lasso penalty uses the $\updownarrow_1$ norm $||\hat{\beta}||_1 = \sum_{j=1}^{p} |\hat{\beta}_j|$, whereas ridge regression uses the square $\updownarrow_2$ norm $||\hat{\beta}||_2^2 = \sum_{j=1}^{p} \beta_j^2$.

Principal component regression. *Principal component analysis* finds the directions in a data set $x_1, ..., x_n \in \mathbb{R}^p$ that have the most variance. Equivalently, principal component analysis finds a low-dimensional subspace that reconstructs the data as well as possible. After performing principal component analysis, we can try *principal component regression* (PCR), where we use a small number of principal components (e.g. 1-20) as predictors.

5 Nonlinearity

5.1 Polynomial Regression

<u>Idea #1</u> Replace simple linear regression.

$$y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

with polynomial regression

$$y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + ... + \beta_\alpha X_i^d + \epsilon_i$$

$\boxed{ex}$ Wage vs age

We can separate out "high earners" and run logistic polynomial regression.

$$\mathbb{P}\{y_i > 250|X_i\} = \frac{\exp(\beta_0 + \beta_1 X_i + ... + \beta_\alpha X_i^d)}{1 + \exp(\beta_0 + \beta_1 X_i + ... + \beta_\alpha X_i^d)}$$

<u>Idea #2</u> Split a quantitative variable into an ordered categorical variable.

We have new predictors:

$$\begin{aligned} C_0(X) &= \mathbb{I}\{X < C_1\} \\ C_1(X) &= \mathbb{I}\{c_1 \leq X < c_2\} \\ &\vdots \\ C_{k-1}(X) &= \mathbb{I}\{C_{k-1} \leq X < C_k\} \\ C_k(X) &= \mathbb{I}\{C_k \leq X\} \end{aligned}$$

where $C_1, C_2, ..., C_k$ are cutpoints. Our book uses cutpoints $C_1 = 35, C_2 = 50, C_3 = 85$.

We can then run linear regression with

$$y_i = \beta_0 + \beta_1 C_1(X_i) + \beta_2 C_2(X_i) + ... + \beta_k C_k(X_i) + \epsilon_i$$

in this model, there are k degrees of freedom in this model.

Idea #3 Use piecewise polynomials (typically $d = 3$)

$$y_i - \begin{cases} \beta_{01} + \beta_{11} X_i + \beta_{21} X_i^2 + \beta_{31} X_i^3 + \epsilon_i & X_i < C \\ \beta_{02} + \beta_{12} X_i + \beta_{22} X_i^2 + \beta_{32} X_i^3 + \epsilon_i & X_i \geq C \end{cases}$$

This usually results in discontinuities within the data.

Idea #4 Use cubic splines
A cubic spline is a piecewise cubic model that imposes the continuity of

1. Prediction function
2. Its first derivative
3. Its second derivative

To find the number of free coefficients with k cutpoints / "knots" $\xi_1, \xi_2, ..., \xi_k$, we introduce a new predictor

$$\begin{aligned} h(X_i \xi_i) &= (X - \xi_i)^3 \\ &= \begin{cases} (X - \xi_i)^3 & X > \xi_i \\ 0 & x \leq \xi_i \end{cases} \end{aligned}$$

To fit cubic splines with k knots, we run multiple linear regression with an intercept and predictors $X, X^2, X^3, h(X_i \xi_1), ..., h(X_i \xi_k)$ with $k + 4$ free variables βs.

5.2 Cubic Splines

1. Choose knots $\xi_1, ..., \xi_k$.
   ex: $\xi_1 = 20,\ \xi_2 = 25,\ \xi_3 = 30$, etc.
2. Run linear regression with

- Intercept
- Predictors $X,\ X^2,\ X^3$
- Predictors $h(X, \xi_i) = (X-\xi)_+^3 = \begin{cases} (X-\xi)^3 & X > \xi \\ 0 & \text{otherwise} \end{cases}$

  for $i = 1, ..., k$

3. Resulting function is

$$y(X) = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \sum_{i=1}^{k} \beta_{k+3}(X-\xi)_+^3$$

$$y'(X) = \beta_1 + 2\beta_2 X + 3\beta_3 X^2 + \sum_{i=1}^{k} 3\beta_{k+3}(X-\xi)_+^2$$

$$y''(X) = 2\beta_2 + 6\beta_3 X + \sum_{i=1}^{k} 6\beta_{k+3}(X-\xi)_+$$

$$y'''(X) = 6\beta_3 + \sum_{i=1}^{k} 6\beta_{k+3}\mathbb{I}\{X > \xi\}$$

Terminology. These are sometimes called "regression splines" because they are fit with multiple linear regression.

- There is also "natural splines" which is "regression splines" plus a restriction that the prediction is linear (not cubic) for $X < \xi_1,\ X > \xi_k$. This one has a special basis with an intercept and $k+3$ predictors.

- Last we study "smoothing splines" which find a function g that minimizes

$$\sum_{i=1}^{n}(y_i - g(X_i))^2 + \underbrace{\lambda \int_{-\infty}^{\infty} g''(t)^2 dt}_{\text{tuning/regularization parameter}}$$

  for a data set $(X_i,\ Y_i)_{1\leq i\leq n}$

Smoothing splines. $\lambda = 0 \Rightarrow$ penalty has no effect so we fit the data perfectly.
$\lambda \to \infty \Rightarrow g''(t) = 0$ everywhere $\Rightarrow g(t) = at + b \Rightarrow g$ is the simple linear

regression predictor.
$\lambda \in (0, \infty) \Rightarrow g$ is a natural cubic spline with knots at $X_1, ..., X_n$
In practice, λ controls the level of smoothness vs. wiggliness $\Rightarrow$ tune λ with a validation set or cross-validation.

Generalized additive model. We replace multiple linear regression

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + ... + \beta_p X_{ip} + \epsilon_i$$

with

$$y_i = \beta_0 + \underbrace{f_1(X_{i1})}_{\text{nonlinear function}} + f_2(X_{i2}) + ... + f_p(X_{ip}) + \epsilon_i$$

There's a cool approach called "backfitting" where we learn each function f_i by predicting the "partial residual."

$\boxed{ex}$ $p = 3 \quad \hat{y}_i = \beta_0 + f_1(X_{i1}) + f_2(X_{i2}) + f_3(X_{i3})$
Set $\beta_0 = \bar{y}..$ Set $f_1,\ f_2,\ f_3 = 0$
Repeat until convergence:

1. Train f_1 to predict $y_i - \beta_0 - f_2(X_{i2}) - f_3(X_{i3})$
2. Train f_2 to predict $y_i - \beta_0 - f_1(X_{i1}) - f_3(X_{i3})$
3. Train f_3 to predict $y_i - \beta_0 - f_1(X_{i1}) - f_2(X_{i2})$

5.3 Regression Trees

$\boxed{ex}$ Hitters data. Predict salary by

1. Log transform data
2. Build tree

Recursive binary splitting.
Split each cell by finding the predictor X_j and cutpoint s that minimizes

$$\sum_{i:X_{ij}<s} (Y_i - \hat{Y}_{R_1})^2 + \sum_{i:X_{ij}\geq s} (Y_i - \hat{Y}_{R_2})^2$$

where $\hat{Y}_{R_1}$ is mean response for predictors $X_{ij} < S$ and $\hat{Y}_{R_2}$ is mean response for predictors $X_{ij} \geq S$

Rectangles

$$R_1 = \{X = \begin{bmatrix} x_1 \\ \vdots \\ x_p \end{bmatrix} \mid x_j < s\}$$

Weakest link pruning.
Given a big tree T_0, find a subtree (T_0 with some branches removed) $T \subseteq T_0$ that minimizes

$$\sum_{m=1}^{|T|} \sum_{i:X_i \in R_m} (y_i - \hat{y}_{R_m})^2 + \alpha|T|$$

where R_m is a rectangle corresponding to a leaf, α is the penalty term, and $|T|$ is the number of leaves.
As we increase α from 0 to ∞ branches get pruned in a nested manner.

Classification tree.
Compare

$$f(x) = \beta_0 + \sum_{j=1}^{p} \beta_j X_j$$

vs.

$$f(x) = \sum_{m=1}^{M} c_m \mathbb{I}\{X \in R_m\}$$

Bagging.
Create B bootstrap data sets. Train decision trees. Average predictions.

$$\hat{f}_{\text{bag}}(X) = \frac{1}{B} \sum_{b=1}^{B} \hat{f}^{(b)}(X)$$

Random Forest.
Create B bootstrap data sets. For each data set, uniformly randomly choose

$\approx \sqrt{p}$ predictors and train a decision tree using only chosen predictors + bootstrap data. Average predictions as before

$$\hat{f}(X) = \frac{1}{B}\sum_{b=1}^{B} \hat{f}^{(b)}(X)$$

Works a little better than bagging.

5.4 Maximum Margin Classifiers

<u>Question</u>: what is a hyperplane?

In 2D, it is a line, a set of vectors $X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ such that $\beta_0 + \beta_1 X_1 + \beta_2 X_2 = 0$. Technically, this is an "affine" hyperplane since it doesn't necessarily pass through (0,0).

In p dimensions, a hyperplane is a set of vectors $X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{bmatrix}$ such that $\beta_0 + \beta_1 X_1 + \beta_2 X_2 + ... + \beta_p X_p$.

<u>Question</u>: how far away is the origin from the hyperplane?

Divide through by $(\sum_{j=1}^{p} \beta_j^2)^{\frac{1}{2}} = \left\| \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_p \end{bmatrix} \right\|$

$\boxed{ex}$ $1 + 2X_1 + 3X_2 = 0$

Divide through by $\sqrt{13}$, we get the following hyperplane equation

$$\frac{1}{\sqrt{13}} + \frac{2}{\sqrt{13}}X_1 + \frac{3}{\sqrt{13}}X_2 = 0$$

For any point $X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, $\frac{1}{\sqrt{13}} + \frac{2}{\sqrt{13}}X_1 + \frac{3}{\sqrt{13}}X_2$ is the "signed distance" to the hyperplane.

<u>Question</u>: what is the distance from $X = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$ to the hyperplane?

Signed distance = $\frac{-4}{\sqrt{13}}$, which means distance = $\frac{4}{\sqrt{13}}$

- If $\beta_0 + \beta_1 X_1 + ... + \beta_p X_p > 0$ you lie on one side of the hyperplane.
- If $\beta_0 + \beta_1 X_1 + ... + \beta_p X_p < 0$ you lie on the opposite side of the hyperplane.
- If it works perfectly to separate data, this is called a "separating hyperplane."

We are given observations $(X_i, y_i)_{1 \le i \le n}$ where $y_i \in \{-1, +1\}$ (any binary outcome). We want a separating hyperplane.

$$\begin{cases} \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + ... + \beta_p X_{ip} > 0 & \text{if } y_i = +1 \\ \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + ... + \beta_p X_{ip} < 0 & \text{if } y_i = -1 \end{cases}$$

Equivalently,

$$y_i(\beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + ... + \beta_p X_{ip}) > 0$$
$$\text{for each } i = 1, 2, ..., n$$

This could be used for out-of-sample predictions

$$\hat{y}_i = \text{sign}(\beta_0 + \beta_1 X_{i1} + ... + \beta_p X_{ip})$$
$$\text{for } i = n+1, n+2, ...$$

Let's write the optimization problem "maximum margin" solves.
Maximize the Euclidean distance from the nearest data points to the hyperplane m such that

$$\begin{cases} \sum_{j=1}^{p} \beta_j^2 = 1 \\ y_i(\beta_0 + \beta_1 X_{i1} + ... + \beta_p X_{ip}) \ge m \end{cases}$$
$$\text{for each } i = 1, ..., n$$

For data in "general position" (ex. with tiny jitter added to data points), there are exactly $p+1$ support vectors that lie a distance m from the hyperplane with at least one on both sides of the hyperplane.

1. Program the optimization and solve it on a computer (no closed form solution).
2. This results in the mathematical properties above.

5.5 Support Vector Classifiers

1. Solve the related optimization problem on your computer.

2. Maximize m such that

$$\begin{cases} \sum_{j=1}^{p} \beta_j^2 = 1 \\ y_i(\beta_0 + \beta_1 X_{i1} + ... + \beta_p X_{ip}) \geq m(1 - \epsilon_i) \quad \text{for } i = 1, ..., n \\ \epsilon_i \geq 0, \quad \sum_{i=1}^{n} \epsilon_i \leq C \leftarrow \text{tuning parameter} \end{cases}$$

- $\epsilon_i > 0$: Observation i violates the margin.
- $\epsilon i > 1$: Observation i lies on the wrong side of the hyperplane.

5.5.1 Support Vector Machines

This method is a flexible non-linear classifier.
Question: How do we do non-linear machine learning?
Idea #1: Add quadratics, cubics, etc.

$$f(x) = \beta_0 + \sum_{j=1}^{p} \beta_{j1} X_j + \sum_{j=1}^{p} \beta_{j2} X_j^2 \quad X = \begin{bmatrix} X_1 \\ \vdots \\ X_p \end{bmatrix}$$

$$\begin{cases} \text{if } f(x) > 0, \text{ guess } \hat{y} = +1 \\ \text{if } f(x) < 0, \text{ guess } \hat{y} = -1 \end{cases}$$

Idea #2: Use the "kernel trick," where we replace every inner product $< x, y >= \sum_{j=1}^{p} x_j y_j$ with a different nonlinear inner product

$$K(x, y) = \exp(-\gamma \sum_{j=1}^{p} (x_j - y_j)^2)$$
$$\text{for } \gamma > 0$$

$\boxed{ex}$ $K(x,y) = (1 + \sum_{j=1}^{p} x_j y_j)^d$
Then we use the prediction function

$$f(x) = \beta_0 + \sum_{i=1}^{n} \alpha_i \exp(-\gamma \sum_{j=1}^{p} (x_j - x_{ij})^2)$$

or

$$f(x) = \beta_0 + \sum_{i=1}^{n} \alpha_i (1 + \sum_{j=1}^{p} x_j, x_{ij})^d$$

or

$$f(x) = \beta_0 + \sum_{i \in S} \alpha_i \exp(-\gamma \sum_{j=1}^{p} |x_j - x_{ij}|)$$